



# DRIVETRAIN TECHNICAL PAPER

## Abstract

The drivetrain system of a formula student car, plays a critical role in the overall performance of the vehicle. This specific paper, presents the calculations, analysis, testing and the final analysis of a drivetrain system developed for a Formula Student car. The vehicle is powered by a Yamaha MT-07 engine. This specific study, dives deep on the selection of the drivetrain system. The study, focuses on a step by step analysis of the appropriate driven gear, chain length and how will the vehicle performs with this specific choices that the team made.

A complete drivetrain model was developed, to evaluate transmission from the engine to the wheels. Gear ratio optimization was performed in order to achieve an effective balance, maximum speed and acceleration of the vehicle while maintaining the engine operation within the optimal torque band. Additionally, sprocket and chain selection were tested based on the maximum performance that this vehicle can give.

## 1. Introduction

Formula Student, is one of the most demanding competitions for engineering university students, because it combines vehicle design, manufacturing and high vehicle performance in a competitive environment. Each team, is faced with the challenge to design and build a race car, fully

capable of competing car, while also must explain to the judges every step and every decision that the team made.

One of the most important vehicle systems, is the drivetrain because it allows the vehicle to move, affects the vehicles' acceleration, power delivery and performance in general. A properly designed drivetrain system, can improve lap times and drivability, while a bad design can affect negatively the vehicles' performance even if the output engine torque is high.

The purpose of this study, is to develop a right drivetrain system witch matches with the MT-07 engine, especially focused on the driven gear, chain and gear ratio selections. This system was developed by using analytical calculations for the chain selection. After that, the team tested different driven gears based on the drivers' needs. Later on, the gear ratio and speed were calculated. Thirdly, by torque curve, a drivetrain analysis took place. Last but not least a kinematics model was developed, and lastly, a force versus speed diagram was made in order to help the driver change the gears effectively based on the whole powertrain study.

## 2. Methodology

The drivetrain system, was developed in order to achieve balance between acceleration, maximum speed, reliability and drivability. The methodology was



# DRIVETRAIN TECHNICAL PAPER

divided in five stages: chain selection, sprocket selection, transmission analysis, kinematic model and tractive force analysis.

## 3. Vehicle Specifications

In the following board, the vehicle specifications are shown. These were the parameters that were served as the foundation of all following analyses and calculations. The following parameters, includes the engine characteristics, vehicle mass and wheel dimensions. The engine characteristics were found from Yamaha MT-07 service manual and the other parameters for the vehicle dynamics team and the chassis team.

| VEICILE SPECS             |                   |
|---------------------------|-------------------|
| Max Torque                | 61 Nm             |
| Chain Type                | 520               |
| Max Engine RPM            | 8.500 (set by us) |
| Vehicle Mass (max)        | 310 Kg            |
| Chain Pitch               | 0.625 in          |
| Center Distance           | 210 mm            |
| Number of Driver Sprocket | 16 Teeth          |
| Safety Factor             | 4                 |

Table 1: Vehicle Specifications.

## 4. Calculating the Theoretical Speed

The first step of the powertrain design, was the analysis of the Yamaha MT-07 top speed per gear per RPM. The provided speed from the manufacturer was the starting point of the first analysis.

The manufacturer speed data were used as a reference during the preliminary drivetrain design process to compare different sprocket combinations. Since the Formula Student vehicle had not yet been completed, the calculations were intended to compare different drivetrain configurations rather than predict the exact vehicle speed.

| Speed at 5000 rpm | MT-07    |
|-------------------|----------|
| 1st               | 25 mph   |
| 2nd               | 33.6 mph |
| 3rd               | 43.7 mph |
| 4th               | 54.8 mph |
| 5th               | 65.4 mph |
| 6th               | 73.9 mph |

Table 2: Speed at 5000 RPM by the Manufacturer.

The next step, was calculating the speed at km/h. The equation was:

$$\frac{km}{h} = mph \cdot 1.60934$$

After the previous calculation, the following top speed at 5,000 RPM is:

| SPEED AT 5,000 RPM PER GEAR |             |
|-----------------------------|-------------|
| 1st                         | 40.125 km/h |
| 2nd                         | 53.9 km/h   |
| 3rd                         | 70.13 km/h  |
| 4th                         | 88 km/h     |
| 5th                         | 105 km/h    |
| 6th                         | 119 km/h    |

Table 3: Theoretical Speed at 5000 RPM.

The third step, was calculating the theoretical speed at 8.500RPM using the Rule of Three after



# DRIVETRAIN TECHNICAL PAPER

converting the original values from mph to km/h.

The equation is:

$$U = U_{5000} \cdot \frac{8500}{5000} = U_{5000} \cdot 1,7$$

After using the equation, the following numbers are the theoretical speed per gear at 8,500 RPM.

| SPEED AT 8.500 RPM |             |
|--------------------|-------------|
| 1st                | 68,2 km/h   |
| 2nd                | 100,81 km/h |
| 3rd                | 119,2 km/h  |
| 4th                | 149,6 km/h  |
| 5th                | 178,5 km/h  |
| 6th                | 202,3 km/h  |

Table 4: Theoretical Speed at 8500 RPM.

## 5. Gear Ratio Testing

After calculating the theoretical top speed, by talking to the driver of the team, the ideal gear ratio and top speed were calculated. For this to happen, the 16T driver sprocket remained the same and the driven sprocket was changed in order to achieve the final result. The manufacturer ratio is 16T/43T.

### Test No1: 16T/52T

In order to calculate the maximum speed with the driven sprocket being 52T, a calculation about the new gear ratio needed to be done first.

The factory gear ratio was:

$$GR_{Before} = \frac{43}{16} = 2.68.$$

The gear ratio after changing the driven sprocket was:

$$GR_{After} = \frac{54}{16} = 3.25.$$

Therefore, the speed reduction factor can be calculated as:

$$\frac{GR_{Before}}{GR_{After}} = \frac{43}{54} \approx 0,8 < 1$$

By increasing the number of teeth on the driven sprocket, the overall gearing becomes shorter.

To determine the new vehicle speeds, a speed variation factor was calculated based on the ratio between the initial and the modified final drive ratios.

The new top speed per gear was calculated by using the following equation:

$$U_{Final} = U_{Old} \cdot 0.8$$

With 0.8 means the relationship between the gear ratio before and after (speed reduction factor). The final result was the following table.

| SPEED AT 8.500 RPM |             |
|--------------------|-------------|
| 1st                | 55,92 km/h  |
| 2nd                | 82,66 km/h  |
| 3rd                | 98.56 km/h  |
| 4th                | 122,6 km/h  |
| 5th                | 146,37 km/h |
| 6th                | 165,8 km/h  |

Table 5: 16T/52T Theoretical Speed Results.

The first test results were considered satisfactory and were reviewed together with the driver. The calculated vehicle speeds



# DRIVETRAIN TECHNICAL PAPER

indicated a good balance between acceleration and maximum speed. However, the configuration was not selected as the final solution.

The team wanted the vehicle to use mainly the first three gears. For this reason, it was decided that the maximum speed in third gear should be above 100 km/h. It should also be noted that powertrain losses from the wheels were not included in the calculations. As a result, the calculated vehicle speeds are theoretical values and were used to compare different gear selections. After discussions, to be sure that the speed in third gear with the losses is over 100 km/h, the 3<sup>rd</sup> gear speed has to be over 105 to 110 km/h.

## Test No2: 12T/42T

The factory gear ratio was:

$$GR_{Before} = \frac{43}{16} = 2.68.$$

The gear ratio after changing the driven and driving sprocket was:

$$GR_{After} = \frac{42}{12} = 3.5.$$

The speed reduction factor calculated as:

$$\frac{GR_{Before}}{GR_{After}} = \frac{2.68}{3.5} \approx 0.76 > 1$$

The new top speed per gear was calculated by using the previous equation:

$$U_{Final} = U_{Old} \cdot 0.76$$

| SPEED AT 8.500 RPM |             |
|--------------------|-------------|
| 1st                | 51,8 km/h   |
| 2nd                | 76,61 km/h  |
| 3rd                | 90,5 km/h   |
| 4th                | 113,69 km/h |
| 5th                | 135,66km/h  |
| 6th                | 153,74 km/h |

Table 6: 12T/42T Theoretical Speed Results.

The testing for driving sprocket with twelve teeth had to stop in this step because there was no gear on the market compatible with the Yamaha MT-07 engine specifications. As a result, the team decided to discontinue the analysis of this specific sprocket and go back to the sixteen teeth which was already on the engine by factory.

## Test No3: 16T/46T

A third drivetrain configuration using a 16-tooth driving sprocket and a 46-tooth driven sprocket was also evaluated. The calculated vehicle speeds at 8500 rpm are presented in the following table.

The factory gear ratio was:

$$GR_{Before} = \frac{43}{16} = 2.68.$$

The gear ratio after changing the driven sprocket is:

$$GR_{After} = \frac{46}{16} = 2.8.$$

By following the previous steps:

$$\frac{GR_{Before}}{GR_{After}} = \frac{43}{46} \approx 0.93 > 1$$

By using the previous calculation:

$$U_{Final} = U_{Old} \cdot 0.93$$



# DRIVETRAIN TECHNICAL PAPER

| SPEED AT 8.500 RPM (km/h) |        |
|---------------------------|--------|
| 1st                       | 63,43  |
| 2nd                       | 93,75  |
| 3rd                       | 110,86 |
| 4th                       | 139,13 |
| 5th                       | 166,01 |
| 6th                       | 188,14 |

Table 7: 16T/46T Theoretical Speed Results.

This test reached a speed of 110.86 km/h in third gear, which was above the target speed of 100 km/h. For this reason, the results were considered satisfactory.

Drivetrain losses were not included in the calculations because no data were available. As a result, the calculated speeds are only theoretical values and the real vehicle speeds will be lower.

After discussion with the driver, this specific gear combination was approved because it reached the team expectations.

After this test was successful, the speeds were converted from km/h to m/s using the following equation:

$$V_{m/s} = \frac{V_{km/h}}{3.6}$$

The final speed is showing in the following table:

| SPEED AT 8.500 RPM (km/h) | m/s   |
|---------------------------|-------|
| 1st                       | 17,62 |
| 2nd                       | 26,04 |
| 3rd                       | 30,79 |
| 4th                       | 38,65 |
| 5th                       | 46,11 |
| 6th                       | 52,26 |

Table 8: Theoretical Speed in km/h an m/s.

## 6. Chain Length

After selecting the final sprocket combination, the required chain length was calculated.

The purpose of this calculation was to determine the number of pitches of 520 chain needed to connect the driving and driven sprockets while maintaining the selected center distance between the shafts.

The equation used was the following:

$$L = \frac{N + n}{2} + \frac{2C}{p} + \left(\frac{N - n}{2\pi}\right)^2 \cdot \frac{p}{C}$$

Where,

**L** = Chain Length in Pitches

**N** = Number of teeth on **large** (Driven) sprocket

**n** = Number of teeth on **small** (Driver) sprocket

**C** = Center to center distance between sprockets

**p** = Pitch of the chain (e.g. 0.25 inches for #25, 0.35 for #35)

By following the test No3, pitch length of 520 chain and adding the center of distance range, the variables are:

|                      |
|----------------------|
| <b>N</b> = 46 teeth  |
| <b>n</b> = 16 teeth  |
| <b>C</b> = 210 mm    |
| <b>p</b> = 15,785 mm |

Table 9: Vehicle Variables.

After replacing the variables with the previous values, the previous equation becomes:



# DRIVETRAIN TECHNICAL PAPER

$$L = \frac{46 + 16}{2} + \frac{2 \cdot 210}{15.785} + \left(\frac{46 - 16}{2\pi}\right)^2 \cdot \frac{15.785}{210}$$

So,

$$L = 48 \text{ pitches}$$

$$\text{Chain Length} = 48 \cdot 15.785 = 920.75 \text{ mm}$$

## 7. Gear Ratio Calculations

After selecting the final sprocket combination, the gear ratio of each gear was calculated.

The purpose of this calculation was to determine the transmission characteristics of the vehicle and verify the gearbox ratios used in the drivetrain analysis.

The equation used, was the following:

$$V = \frac{RPM \cdot C}{GR \cdot FD \cdot 60}$$

Where,

**GR** = Gear ratio

**RPM** = Rounds per minute

**C** = Wheel circumference (m)

**V** = Vehicle speed (m/s)

**FD** = Final drive ratio

The factor 60 is used to convert the engine speed from revolutions per minute (rpm) to revolutions per second.

After substituting the calculated speed values into the previous equation:

$$GR = \frac{RPM \cdot C}{V \cdot FD \cdot 60}$$

The wheel circumference was first calculated using equation:  $C = \pi \cdot D$

$$C = 3.14 \cdot 0.48 = 1.507 \text{ m}$$

After performing the above calculations, the gear ratios of the transmission were determined.

The final results are presented in the following table:

| GEAR | GR FINAL |
|------|----------|
| 1st  | 3,69     |
| 2nd  | 2,50     |
| 3rd  | 2,11     |
| 4th  | 1,68     |
| 5th  | 1,41     |
| 6th  | 1,24     |

Table 10: Final Gear Ratio for Each Gear.

## 8. Drivetrain Analysis

Using the calculated gear ratios, the theoretical vehicle speed was determined for each gear.

The calculations were performed for engine speeds of 5000 rpm and 8500 rpm in order to examine the operating range of the drivetrain.

| SPEED PER RPM PER GEAR |             |           |       |            |
|------------------------|-------------|-----------|-------|------------|
| GEAR                   | SPEED km/h  | SPEED m/s | RPM   | GEAR RATIO |
| 1st                    | 32,1 km/h   | 8,91 m/s  | 5.000 | 4,23       |
| 2nd                    | 43,12 km/h  | 11,97 m/s | 5.000 | 3,15       |
| 3rd                    | 56,1 km/h   | 15,58 m/s | 5.000 | 2,42       |
| 4th                    | 70,4 km/h   | 19,55 m/s | 5.000 | 1,93       |
| 1st                    | 54,56 km/h  | 15,5 m/s  | 8.500 | 4,23       |
| 2nd                    | 73,3 km/h   | 20,36 m/s | 8.500 | 3,15       |
| 3rd                    | 95,36 km/h  | 26,48 m/s | 8.500 | 2,42       |
| 4th                    | 119,68 km/h | 33,24 m/s | 8.500 | 1,93       |

Table 11: Theoretical Speed Per Gear at 5000 and 8500 RPM.

The angular velocity of the engine was calculated for each engine speed using the following equation:

$$\omega_i = 2\pi \cdot \frac{RPM}{60}$$

Where,

$\omega_i$  = Angular velocity (rad/s)



# DRIVETRAIN TECHNICAL PAPER

**RPM** = Engine speed (rpm)

For an engine speed of 1000 rpm:

$$\omega_i = 2\pi \cdot \frac{1000}{60} = 104.71 \text{ rad/s}$$

The same procedure was repeated for all engine speeds used in this study. The calculated values are presented in the following table.

| <b>ω per RPM</b> |                    |
|------------------|--------------------|
| <b>RPM</b>       | <b>ω (rad/sec)</b> |
| 1000             | 104,71             |
| 2000             | 209,43             |
| 3000             | 314,16             |
| 4000             | 418,17             |
| 5000             | 523,6              |
| 6000             | 628,31             |
| 7000             | 733,03             |
| 8000             | 837,75             |
| 8500             | 890,11             |

**Table 12: Theoretical Angular Velocity Per Gear.**

As expected, the angular velocity increases linearly with engine speed.

After calculating the engine angular velocity, the input power was determined using the engine torque values and the following equation:

$$P_i = T_i \cdot \omega_i$$

where:

$P_i$  = input power (W)

$T_i$  = engine torque (Nm)

$\omega_i$  = angular velocity (rad/s)

For the engine torque, an analysis from Riccardo Wave on the MT-07 engine was used. This specific analysis gave the following results:

| <b>TORQUE VS RPM</b> |                    |
|----------------------|--------------------|
| <b>RPM</b>           | <b>TORQUE (Nm)</b> |
| 0                    | 0                  |
| 1000                 | 14                 |
| 2000                 | 24                 |
| 3000                 | 33                 |
| 4000                 | 38                 |
| 5000                 | 47                 |
| 6000                 | 60                 |
| 7000                 | 50                 |
| 8000                 | 42                 |
| 8500                 | 38                 |
| 9000                 | 32                 |
| 10000                | 23                 |

**Table 13: Theoretical Torque VS RPM.**

As an example, for an engine speed of 1000 rpm:

$$P_i = 14 \cdot 104.71 = 1465.94 \text{ W}$$

The same procedure was followed for all engine speeds. The results are presented in the following table.

| <b>INPUT POWER</b> |           | <b>ω per RPM</b> |                    |           |
|--------------------|-----------|------------------|--------------------|-----------|
| <b>RPM</b>         | <b>Ti</b> | <b>RPM</b>       | <b>ω (rad/sec)</b> | <b>Pi</b> |
| 1000               | 14        | 1000             | 104,71             | 1465,94   |
| 2000               | 24        | 2000             | 209,43             | 5026,32   |
| 3000               | 33        | 3000             | 314,16             | 10367,28  |
| 4000               | 38        | 4000             | 418,17             | 15890,46  |
| 5000               | 47        | 5000             | 523,6              | 4921,37   |
| 6000               | 60        | 6000             | 628,31             | 37698,6   |
| 7000               | 50        | 7000             | 733,03             | 36651,5   |
| 8000               | 42        | 8000             | 837,75             | 35185,5   |
| 8500               | 38        | 8500             | 890,11             | 33824,18  |

**Table 14: Final Table of Input Power and Angular Velocity Per RPM (Theoretical).**

The output angular velocity was calculated for each gear ratio and gear, by using the following equation:

$$\omega_o = \frac{\omega_i}{FD \cdot GR}$$

Where,

$\omega_o$  = Output angular velocity (rad/s)



# DRIVETRAIN TECHNICAL PAPER

$\omega_i$  = Input angular velocity (rad/s)

GR = Gear ratio

FD = Final drive ratio

As an example, for first gear at 1000 rpm:

$$\omega_o = \frac{104.71}{3.69 \cdot 2.8} = 9.87 \text{ rad/s}$$

The same procedure was followed for all engine speeds, gears and gear ratios. The calculated values are presented in the following tables.

| <b><math>\omega</math> out per RPM (1st gear)</b> |                                      |
|---------------------------------------------------|--------------------------------------|
| <b>RPM</b>                                        | <b><math>\omega</math> (rad/sec)</b> |
| 1000                                              | 9,87                                 |
| 2000                                              | 19,74                                |
| 3000                                              | 29,61                                |
| 4000                                              | 39,42                                |
| 5000                                              | 49,36                                |
| 6000                                              | 59,23                                |
| 7000                                              | 69,10                                |
| 8000                                              | 78,97                                |
| 8500                                              | 83,90                                |

Table 15: Theoretical Output Angular Velocity in 1st Gear.

| <b><math>\omega</math> out per RPM (2nd gear)</b> |                                      |
|---------------------------------------------------|--------------------------------------|
| <b>RPM</b>                                        | <b><math>\omega</math> (rad/sec)</b> |
| 1000                                              | 14,58                                |
| 2000                                              | 29,17                                |
| 3000                                              | 43,75                                |
| 4000                                              | 58,24                                |
| 5000                                              | 72,92                                |
| 6000                                              | 87,51                                |
| 7000                                              | 102,09                               |
| 8000                                              | 116,68                               |
| 8500                                              | 123,97                               |

Table 16: Theoretical Output Angular Velocity in 2nd Gear.

| <b><math>\omega</math> out per RPM (3rd gear)</b> |                                      |
|---------------------------------------------------|--------------------------------------|
| <b>RPM</b>                                        | <b><math>\omega</math> (rad/sec)</b> |
| 1000                                              | 17,28                                |
| 2000                                              | 34,56                                |
| 3000                                              | 51,84                                |
| 4000                                              | 69,00                                |
| 5000                                              | 86,40                                |
| 6000                                              | 103,68                               |
| 7000                                              | 120,96                               |
| 8000                                              | 138,24                               |
| 8500                                              | 146,88                               |

Table 17: Theoretical Output Angular Velocity in 3rd Gear.

| <b><math>\omega</math> out per RPM (4th gear)</b> |                                      |
|---------------------------------------------------|--------------------------------------|
| <b>RPM</b>                                        | <b><math>\omega</math> (rad/sec)</b> |
| 1000                                              | 21,68                                |
| 2000                                              | 43,36                                |
| 3000                                              | 65,04                                |
| 4000                                              | 86,58                                |
| 5000                                              | 108,41                               |
| 6000                                              | 130,08                               |
| 7000                                              | 151,77                               |
| 8000                                              | 173,45                               |
| 8500                                              | 184,29                               |

Table 18: Theoretical Output Angular Velocity in 4th Gear.

The calculations were carried out up to 8500 rpm and fourth gear.

Based on discussions with the driver, higher engine speeds and gears above fourth are not expected to be used during Formula Student events.

After calculating the output angular velocity, the transmission output torque had to be calculated by using the following calculation:

$$T_o = T_i \cdot GR \cdot FD \cdot \eta$$

In this study, drivetrain losses were not considered due to the lack of experimental data. Therefore, a



# DRIVETRAIN TECHNICAL PAPER

drivetrain efficiency of  $\eta=1$  was assumed.

After calculating for four gears per a thousand RPM and in theoretical vehicle maximum RPM, the results are shown in the following tables:

| OUTPUT TORQUE |        | $\omega$ out per RPM (1st gear) |                    |
|---------------|--------|---------------------------------|--------------------|
| RPM           | To     | RPM                             | $\omega$ (rad/sec) |
| 1000          | 148.52 | 1000                            | 9.87               |
| 2000          | 254.61 | 2000                            | 19.74              |
| 3000          | 350.09 | 3000                            | 29.61              |
| 4000          | 403.13 | 4000                            | 39.42              |
| 5000          | 498.61 | 5000                            | 49.36              |
| 6000          | 636.53 | 6000                            | 59.23              |
| 7000          | 530.44 | 7000                            | 69.10              |
| 8000          | 445.57 | 8000                            | 78.97              |
| 8500          | 403.13 | 8500                            | 83.90              |

**Table 19: Theoretical Transmission Torque and Output Angular Velocity Per RMP in First Gear.**

| OUTPUT TORQUE |        | $\omega$ out per RPM (2nd gear) |                    |
|---------------|--------|---------------------------------|--------------------|
| RPM           | To     | RPM                             | $\omega$ (rad/sec) |
| 1000          | 100.52 | 1000                            | 14.58              |
| 2000          | 172.32 | 2000                            | 29.17              |
| 3000          | 236.94 | 3000                            | 43.75              |
| 4000          | 272.84 | 4000                            | 58.24              |
| 5000          | 337.46 | 5000                            | 72.92              |
| 6000          | 430.80 | 6000                            | 87.51              |
| 7000          | 359.00 | 7000                            | 102.09             |
| 8000          | 301.56 | 8000                            | 116.68             |
| 8500          | 272.84 | 8500                            | 123.97             |

**Table 20: Theoretical Transmission Torque and Output Angular Velocity in Second Gear.**

| OUTPUT TORQUE |        | $\omega$ out per RPM (3rd gear) |                    |
|---------------|--------|---------------------------------|--------------------|
| RPM           | To     | RPM                             | $\omega$ (rad/sec) |
| 1000          | 84.84  | 1000                            | 17.28              |
| 2000          | 145.44 | 2000                            | 34.56              |
| 3000          | 199.98 | 3000                            | 51.84              |
| 4000          | 230.28 | 4000                            | 69.00              |
| 5000          | 284.82 | 5000                            | 86.40              |
| 6000          | 363.60 | 6000                            | 103.68             |
| 7000          | 303.00 | 7000                            | 120.96             |
| 8000          | 254.52 | 8000                            | 138.24             |
| 8500          | 230.28 | 8500                            | 146.88             |

**Table 21: Theoretical Transmission Torque and Output Angular Velocity in Third Gear.**

| OUTPUT TORQUE |        | $\omega$ out per RPM (4th gear) |                    |
|---------------|--------|---------------------------------|--------------------|
| RPM           | To     | RPM                             | $\omega$ (rad/sec) |
| 1000          | 67.62  | 1000                            | 21.68              |
| 2000          | 115.92 | 2000                            | 43.36              |
| 3000          | 159.39 | 3000                            | 65.04              |
| 4000          | 183.54 | 4000                            | 86.58              |
| 5000          | 227.01 | 5000                            | 108.41             |
| 6000          | 289.80 | 6000                            | 130.08             |
| 7000          | 241.50 | 7000                            | 151.77             |
| 8000          | 202.86 | 8000                            | 173.45             |
| 8500          | 183.54 | 8500                            | 184.29             |

**Table 22: Theoretical Transmission Torque and Output Angular Velocity in Fourth Gear.**

The transmission output power was calculated using the output torque and output angular velocity. The following equation was used for all engine speeds and gears.

$$P_o = T_o \cdot \omega_o$$

Where,

$P_o$  = Transmission output power (W)

Drivetrain losses were not considered. Therefore, the calculated output power is equal to the input power.

| OUTPUT TORQUE |        | $\omega$ out per RPM (1st gear) |                    | $P_o$    |
|---------------|--------|---------------------------------|--------------------|----------|
| RPM           | To     | RPM                             | $\omega$ (rad/sec) |          |
| 1000          | 148.52 | 1000                            | 9.87               | 1465.94  |
| 2000          | 254.61 | 2000                            | 19.74              | 5026.32  |
| 3000          | 350.09 | 3000                            | 29.61              | 10367.28 |
| 4000          | 403.13 | 4000                            | 39.42              | 15890.46 |
| 5000          | 498.61 | 5000                            | 49.36              | 24609.20 |
| 6000          | 636.53 | 6000                            | 59.23              | 37698.60 |
| 7000          | 530.44 | 7000                            | 69.10              | 36651.50 |
| 8000          | 445.57 | 8000                            | 78.97              | 35185.50 |
| 8500          | 403.13 | 8500                            | 83.90              | 33824.18 |

**Table 23: Theoretical Output Transmission Power, Transmission Torque and Output Angular Velocity Per RPM in First Gear.**

The output power is the same for all four gears.

## 9. Comparison of Input and Output Angular Velocity.

The following figure, represent an Excel comparison of the input and output angular velocities for the first four gears. As expected, the output



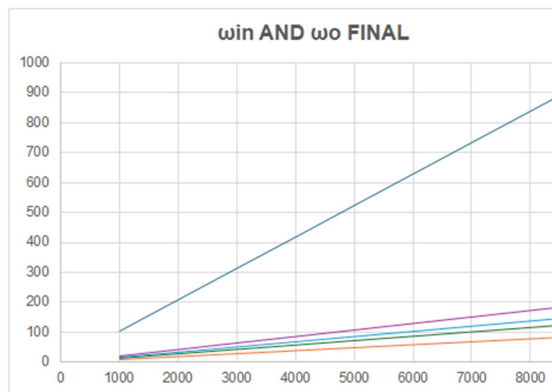
# DRIVETRAIN TECHNICAL PAPER

angular velocity is lower than the input velocity. This happens because of the transmission reduction. The smallest output velocity is seen in the first gear, while the highest in the last.

| win AND wo FINAL |        |        |        |        |        |
|------------------|--------|--------|--------|--------|--------|
| RPM              | win    | wo 1st | wo 2nd | wo 3rd | wo 4th |
| 1000             | 104,71 | 9,87   | 14,58  | 17,28  | 21,68  |
| 2000             | 209,43 | 19,74  | 29,17  | 34,56  | 43,36  |
| 3000             | 314,16 | 29,61  | 43,75  | 51,84  | 65,04  |
| 4000             | 418,17 | 39,42  | 58,24  | 69,00  | 86,58  |
| 5000             | 523,6  | 49,36  | 72,92  | 86,40  | 108,41 |
| 6000             | 628,31 | 59,23  | 87,51  | 103,68 | 130,08 |
| 7000             | 733,03 | 69,10  | 102,09 | 120,96 | 151,77 |
| 8000             | 837,75 | 78,97  | 116,68 | 138,24 | 173,45 |
| 8500             | 890,11 | 83,90  | 123,97 | 146,88 | 184,29 |

**Table 24: Theoretical Input and Output Angular Velocity of All Gears Per RPM.**

The calculated values were then combined to produce the following diagram. The yellow line represents the first gear, green line is the second gear, third gear is represented by blue line and fourth by purple.



**Figure 1: Diagram of Theoretical Input and Output Angular Velocity of All Gears Per RPM.**

The theoretical vehicle speed was calculated from the output angular velocity and the wheel radius. The calculated speed in m/s was then converted to km/h using the conversion factor of 3.6.

The equation is:

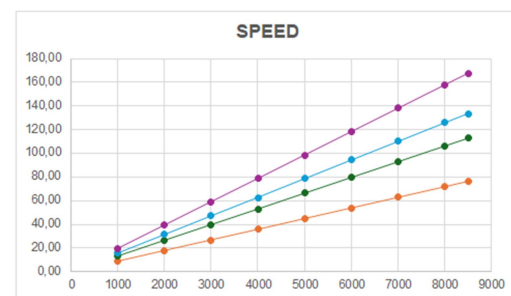
$$V_{km/h} = \omega_o \cdot r_w \cdot 3.6$$

The theoretical speed, without losses, is shown at the following table:

| SPEED |        |       |        |        |        |
|-------|--------|-------|--------|--------|--------|
| RPM   | win    | V 1st | V 2nd  | V 3rd  | V 4th  |
| 1000  | 104,71 | 8,99  | 13,28  | 15,74  | 19,75  |
| 2000  | 209,43 | 17,98 | 26,57  | 31,48  | 39,49  |
| 3000  | 314,16 | 26,97 | 39,85  | 47,22  | 59,24  |
| 4000  | 418,17 | 35,90 | 53,05  | 62,85  | 78,85  |
| 5000  | 523,6  | 44,95 | 66,42  | 78,70  | 98,74  |
| 6000  | 628,31 | 53,94 | 79,70  | 94,43  | 118,48 |
| 7000  | 733,03 | 62,93 | 92,99  | 110,17 | 138,23 |
| 8000  | 837,75 | 71,92 | 106,27 | 125,91 | 157,98 |
| 8500  | 890,11 | 76,42 | 112,91 | 133,78 | 167,85 |

**Table 25: Theoretical Speed of All Gears and Input Angular Velocity Per RPM.**

The maximum speed, is expected to be lower because of the losses and because the manufacturer data were for an MT-07 motorbike. Then, an Excel diagram was made in order to visualize the top speed per gear. The yellow line represents the first gear, the second is represented by the green line, the third by the blue and the fourth by the colour purple.



**Figure 2: Theoretical Maximum Speed Diagram.**

Since the Formula Student vehicle had not yet been completed, the calculated vehicle speeds represent theoretical design values used for drivetrain selection rather than actual vehicle performance.



# DRIVETRAIN TECHNICAL PAPER

## 10. Tractive Force Calculation

The tractive force, was calculated by using the transmission torque and the wheel radius. The calculation was used for evaluation of the force available at the tire and road contact for each gear.

The force versus speed also provides useful information about the operating range of each gear. In simple words, it can also be used as a guide for gear shifting as it shows the speed range where each gear provides the highest tractive force. Drivetrain losses were neglecting because of lack of experiment data.

The tractive force, was calculated by using the wheel torque and radius by the following equation:

$$F = \frac{T_o}{R_w}$$

Where,

F = Tractive force (N)

For the first gear, the tractive force is shown on the following table:

| WHEEL TORQUE PER RPM PER GEAR (1st) |        |         |
|-------------------------------------|--------|---------|
| To = Tw                             | Rwheel | F       |
| 148,52                              | 0,25   | 587,05  |
| 254,61                              |        | 1006,36 |
| 350,09                              |        | 1383,75 |
| 403,13                              |        | 1593,41 |
| 498,61                              |        | 1970,80 |
| 636,53                              |        | 2515,91 |
| 530,44                              |        | 2096,59 |
| 445,57                              |        | 1761,14 |
| 403,13                              |        | 1593,41 |

**Table 26: Theoretical Wheel Torque and Tractive Force in First Gear.**

The second gear tractive force was found to be the following:

| WHEEL TORQUE PER RPM PER GEAR (2nd) |        |         |
|-------------------------------------|--------|---------|
| To = Tw                             | Rwheel | F       |
| 100,52                              | 0,25   | 397,31  |
| 172,32                              |        | 681,11  |
| 236,94                              |        | 936,52  |
| 272,84                              |        | 1078,42 |
| 337,46                              |        | 1333,83 |
| 430,80                              |        | 1702,77 |
| 359,00                              |        | 1418,97 |
| 301,56                              |        | 1191,94 |
| 272,84                              |        | 1078,42 |

**Table 27: Theoretical Wheel Torque and Force in Second Gear.**

Then, the tractive force of the third gear was calculated:

| WHEEL TORQUE PER RPM PER GEAR (3rd) |        |         |
|-------------------------------------|--------|---------|
| To = Tw                             | Rwheel | F       |
| 84,84                               | 0,25   | 335,34  |
| 145,44                              |        | 574,86  |
| 199,98                              |        | 790,43  |
| 230,28                              |        | 910,20  |
| 284,82                              |        | 1125,77 |
| 363,60                              |        | 1437,15 |
| 303,00                              |        | 1197,63 |
| 254,52                              |        | 1006,01 |
| 230,28                              |        | 910,20  |

**Table 28: Theoretical Wheel Torque and Force in Third Gear.**

Lastly, the final table represents the fourth gear's tractive force.

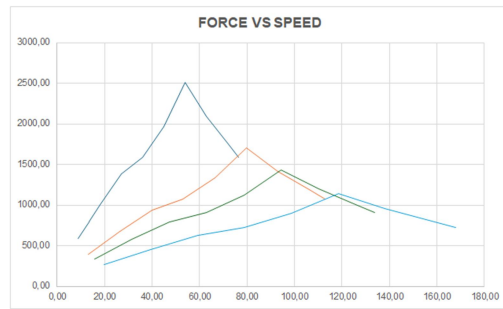
| WHEEL TORQUE PER RPM PER GEAR (4th) |        |         |
|-------------------------------------|--------|---------|
| To = Tw                             | Rwheel | F       |
| 67,62                               | 0,25   | 267,27  |
| 115,92                              |        | 458,18  |
| 159,39                              |        | 630,00  |
| 183,54                              |        | 725,45  |
| 227,01                              |        | 897,27  |
| 289,80                              |        | 1145,45 |
| 241,50                              |        | 954,55  |
| 202,86                              |        | 801,82  |
| 183,54                              |        | 725,45  |

**Table 29: Theoretical Wheel Torque and Force in Fourth Gear.**

After all the previous tables were created, were converted to an Excel diagram which also guides the driver about gear shifting.



# DRIVETRAIN TECHNICAL PAPER



**Figure 3: Theoretical Force VS Speed Diagram.**

## 11. Results and Discussion

Three different sprocket combinations were evaluated during the drivetrain design process. The first combination (16/52) provided high acceleration but reduced the maximum vehicle speed. The second combination (12/42) was not proceeded for further investigation because a suitable 12 teeth gear was not available at the market. The final combination (16/46), provided the best choice between the previous testings because the third gear exceeded 100 km/h in theory, which satisfied the team and the driver. Although calculated speeds are theoretical because the lack of drivetrain losses, the selected drivetrain meets the required performance for Formula Students events.

The drivetrain analysis showed that the selected transmission provides satisfactory tractive force at low vehicle speeds. As expected, the tractive force decreases as the vehicle speed increases.

## 12. Conclusions

This study presented the design and analysis of a drivetrain system for a Formula Student car, powered by a Yamaha MT-07 engine. A systematic methodology was followed, including the testing of different sprocket combinations, chain and gear ration calculations, power transmission analysis and tractive force calculations.

Three sprocket combinations were investigated with the final decision being a 16/46 sprocket ratio. The drivetrain analysis showed that this specific selection provides a good balance between speed and acceleration.

Although drivetrain losses were not included in the calculations, this study provides the base of the powertrain design which will be used in the team's Formula Student car. The results can be used to support future drivetrain optimization and vehicle performance improvements.